

QUANTUM SCATTERING REPORT

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0. Model

The numerical calculations use the lecture convention $m=1$ and $\hbar=1$. For the free Gaussian,

$$\psi(0,x) = (2\alpha/\pi)^{1/4} \exp[-\alpha(x-x_0)^2] \exp[ip_0(x-x_0)]$$

The interacting calculations use

$$V(x) = (0.5x^2 - 0.8) \exp(-0.1x^2).$$

This is a symmetric central well surrounded by two barriers.

1. Part 1 - Free Gaussian wavepacket

1.1 Initial Gaussian: changing α

At fixed mean momentum $p_0=1$, increasing α decreases the width of the wavepacket in real space, since $\Delta x=1/(2\sqrt{\alpha})$. Therefore, the Gaussian becomes narrower as α increases.

initial gaussians - width is changing

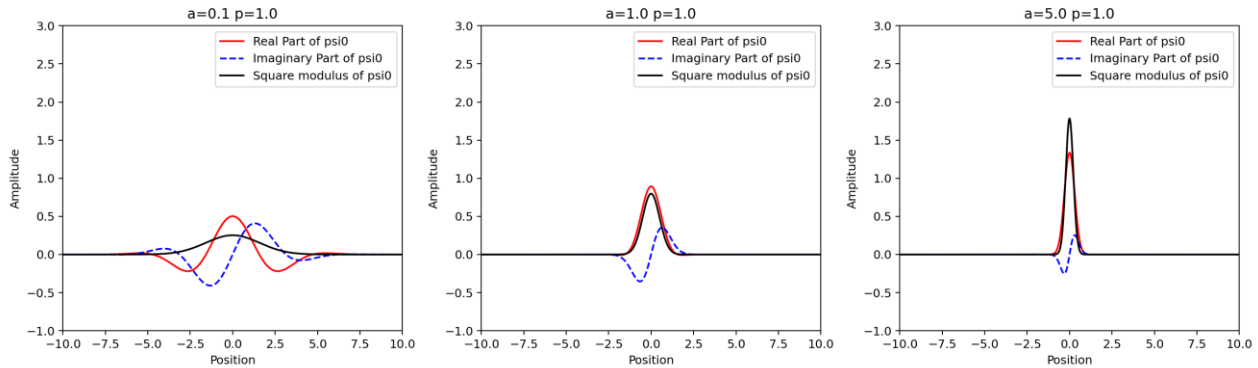


Figure 1. Initial Gaussian states for $\alpha=0.1, 1$, and 5 at fixed $p_0=1$. The black curve is $|\psi|^2$; red/blue curves are the real/imaginary parts.

1.2 Initial Gaussian: changing mean momentum p_0

At fixed $\alpha=1$, as p_0 increases, the wavefunction oscillates more rapidly, while the probability density $|\psi|^2$ remains unchanged.

initial gaussians - momentum is changing

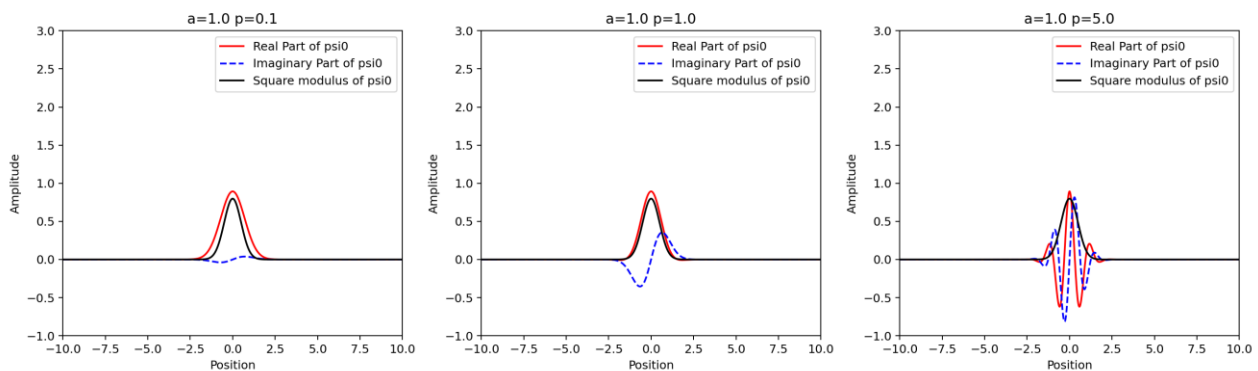


Figure 2. Initial Gaussian states for $p_0=0.1, 1$, and 5 at fixed $\alpha=1$. The phase oscillates more rapidly as p_0 increases.

1.3 Free propagation and spreading (t>0)

As time increases, the center of the free Gaussian wavepacket moves to the right with velocity p_0/m . At the same time, the wavepacket spreads in real space, so its spatial width increases continuously during propagation.

free wavepacket propagation - snapshots

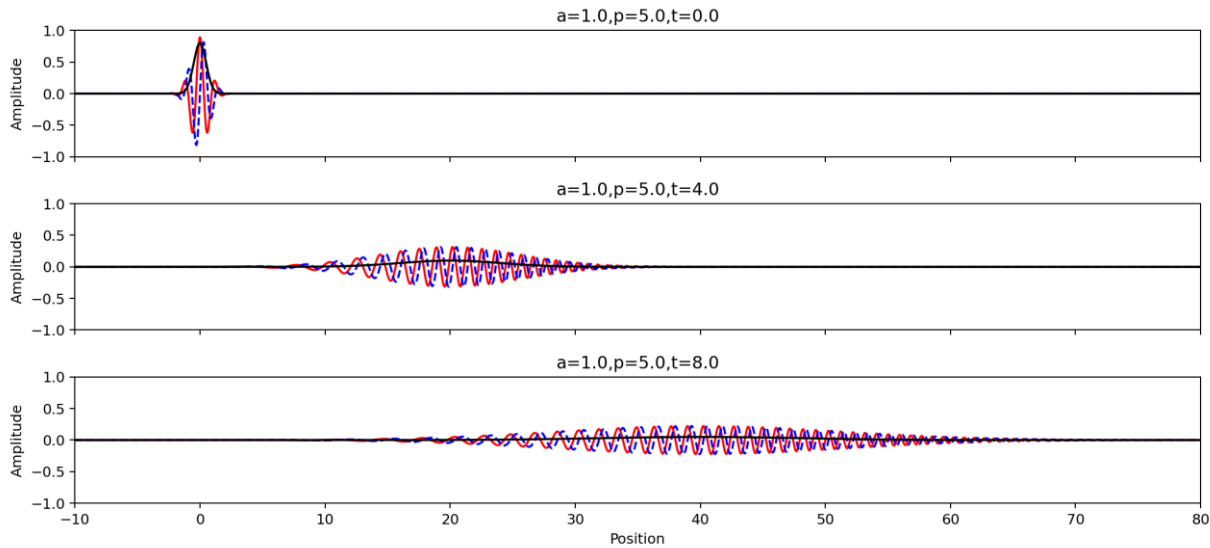


Figure 3. Free propagation snapshots for $\alpha=1$, $p_0=5$. The wavepacket moves to the right at the expected group velocity and broadens with time.

2. Part 2 - Regularized Dirac delta function

The regularization used in the project is

$$\delta_L(p) = \sin(pL)/(\pi p), \quad \delta_L(0)=L/\pi.$$

As L increases, the central peak becomes taller and narrower. Although $\delta_L(p)$ does not converge pointwise, it approaches the Dirac delta distribution in the sense that $\int \varphi(p) \delta_L(p) dp \rightarrow \varphi(0)$.

2.1 Numerical distribution test

The test function is the momentum-space Gaussian $\varphi(p)$ with $\alpha=1$ and $p_0=5$. The numerical integration window is fixed at $p \in [-10, 10]$ for every L .

L	$\int \varphi(p) \delta_L(p) dp$	$\varphi(0)$	Absolute difference
10	0.0012166507	0.0012193111	2.660e-06
100	0.0012190874	0.0012193111	2.237e-07
1000	0.0012193480	0.0012193111	3.696e-08

The error decreases from 2.66×10^{-6} at $L=10$ to 3.70×10^{-8} at $L=1000$. The convergence toward $\varphi(0)=0.0012193111$ is a direct numerical verification of the defining action of the Dirac delta distribution.

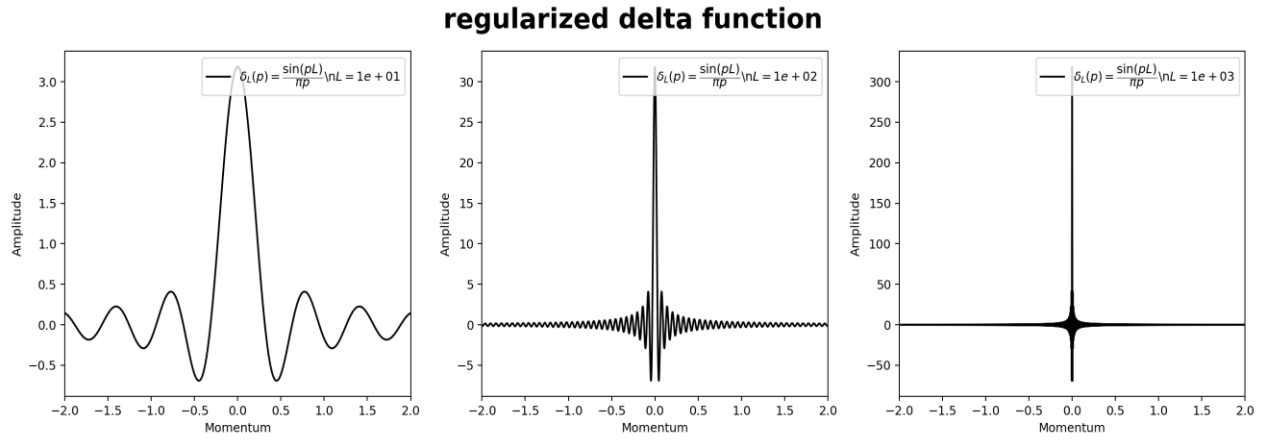


Figure 4. The regularized delta function for $L=10, 100$, and 1000 . Increasing L concentrates the distribution more strongly around $p=0$.

3. Part 3 - Stationary scattering and resonances

3.1 Numerical method

For each $E>0$, let $k=\sqrt{2E}$ under $m=\hbar=1$. The physical stationary state for a particle incident from the left has the asymptotic form

$$\begin{aligned}\psi_E(x \rightarrow -\infty) &= \exp(ikx) + R(E) \exp(-ikx) \\ \psi_E(x \rightarrow +\infty) &= T(E) \exp(ikx)\end{aligned}$$

For the finite-difference calculation, a modified state with unit transmitted amplitude is used instead:

$$\begin{aligned}\tilde{\psi}_E(x \rightarrow -\infty) &= A(E) \exp(ikx) + B(E) \exp(-ikx) \\ \tilde{\psi}_E(x \rightarrow +\infty) &= \exp(ikx) \\ \psi_E(x) &= \tilde{\psi}_E(x)/A(E), \quad T(E)=1/A(E), \quad R(E)=B(E)/A(E).\end{aligned}$$

The physical transmission and reflection probabilities are $|T|^2$ and $|R|^2$, with $|T|^2+|R|^2=1$ as the numerical consistency check.

3.2 Transmission spectrum

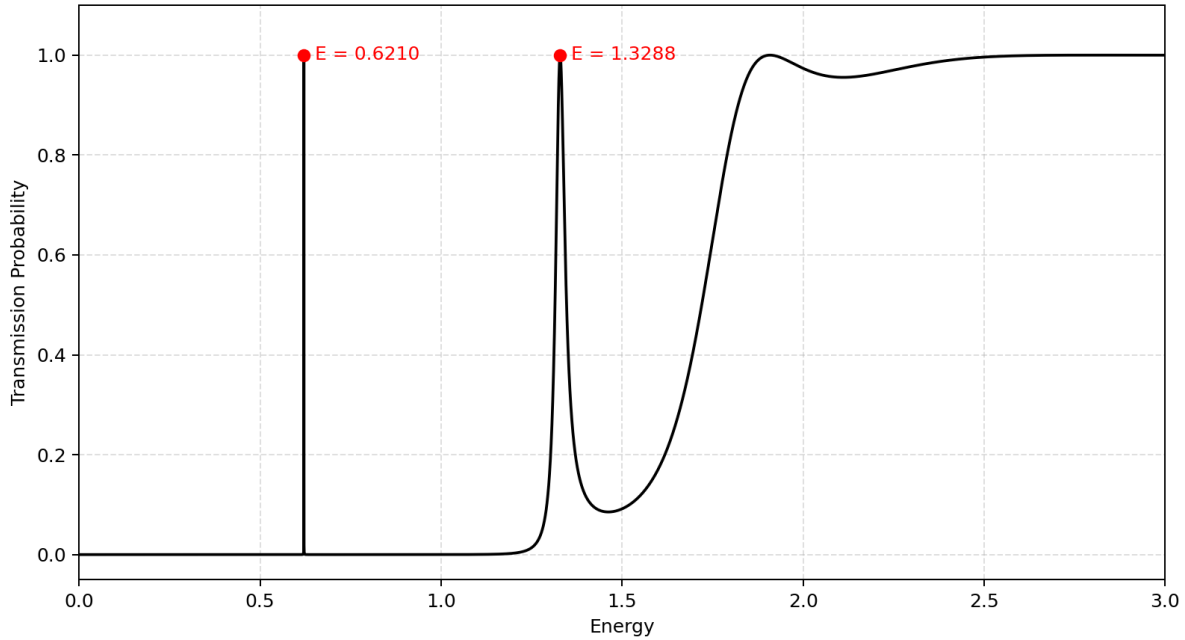


Figure 5. Calculated transmission probability over $0 < E < 3$. Two sharp below-barrier resonances are visible at $E \approx 0.6210$ and $E \approx 1.3288$; at higher energy the transmission tends toward unity.

The transmission spectrum shows two narrow resonance peaks at $E \approx 0.6210$ and $E \approx 1.3288$, where the transmission probability reaches almost one. At higher energies, the transmission gradually approaches one.

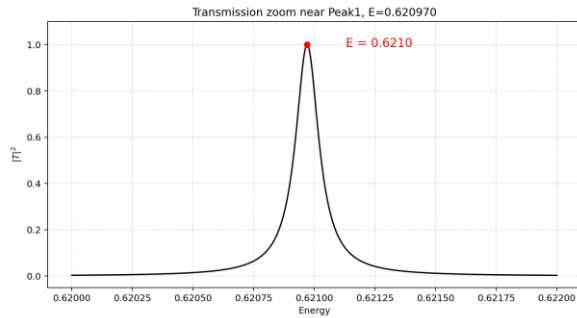


Figure 6a. Zoom of the first peak at $E \approx 0.6210$

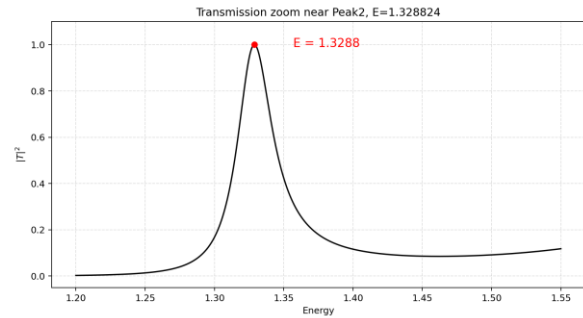
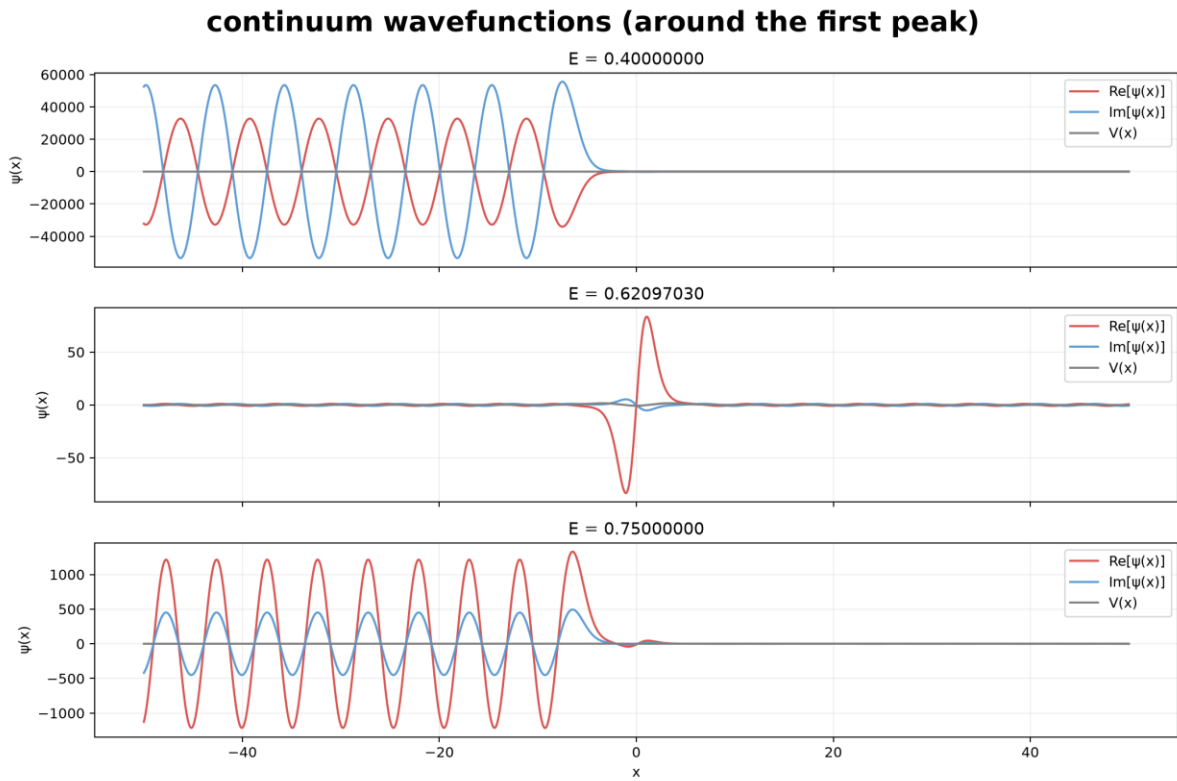


Figure 6b. Zoom of the second peak $E \approx 1.3288$

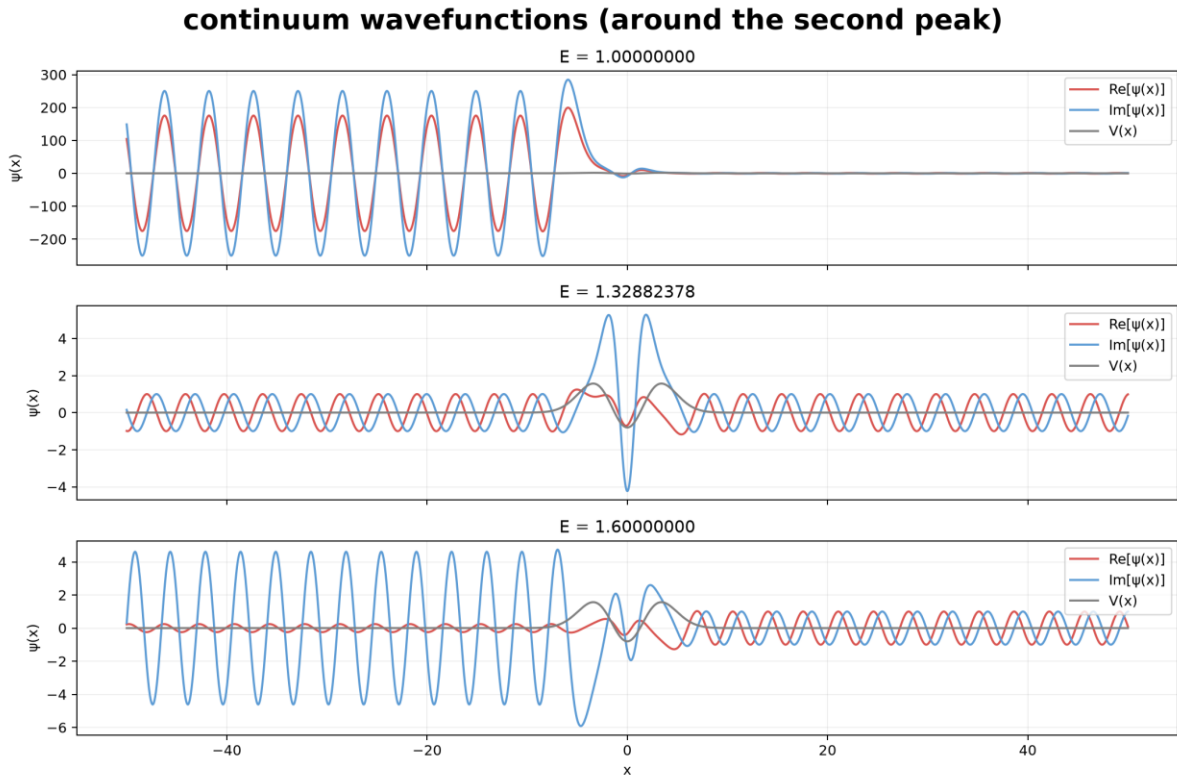
3.3 Continuum wavefunctions around the resonances

The following panels show the modified state $\tilde{\psi}$ defined in Section 3.1.

First resonance. At $E=0.40$ and $E=0.75$, the physical transmission is almost zero and the wave is almost completely reflected. At the resonance energy $E \approx 0.62097$, $|T|^2 \approx 1$ and the wave is almost perfectly transmitted through the potential, and a large oscillation is also formed inside the central well.



Second resonance. The same behavior appears around $E \approx 1.32882$. Away from the peak ($E=1.00$ and $E=1.60$), the transmission is much smaller, whereas at the resonance $|T|^2 \approx 1$ and the wave is almost perfectly transmitted through the potential, and a large oscillation is also formed inside the central well.



4. Part 4 - Wavepacket scattering

In Part 4, a Gaussian wavepacket is used as the incident wavepacket. It is constructed by superposing the left-incident stationary scattering states obtained in Part 3 with a Gaussian momentum distribution:

$$\Psi(t, x) = \int dp \phi(p) \psi_p(x) e^{-iE(p)t}, \quad E = p^2/2.$$

4.1 Blocked packet

The blocked packet is centered at $E_0 = 1.0$, well below the barrier top and away from either resonant peak. Across the relevant momentum distribution the transmission is extremely small. Therefore, almost the entire wavepacket is reflected back to the left. The snapshots show exactly this: the packet approaches the barrier, overlaps the potential well region, and then comes back as a reflected packet with negligible transmitted weight.

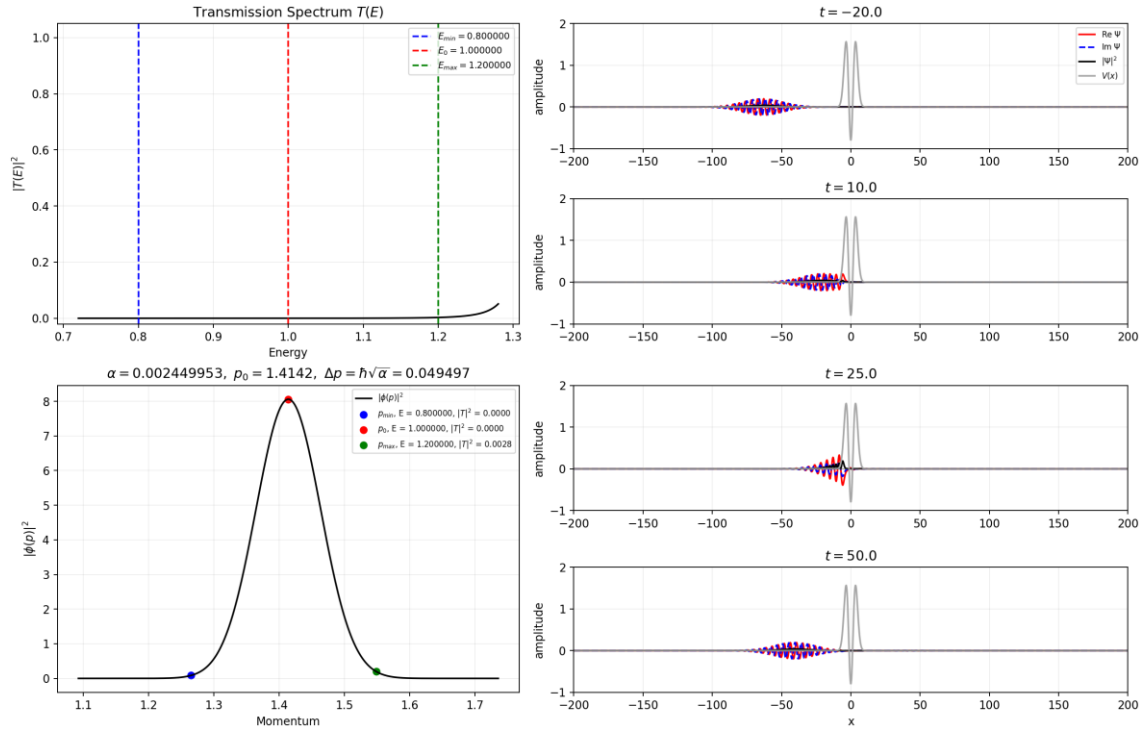


Figure 9. Blocked scenario. Left: stationary transmission spectrum and incident momentum density. Right: time snapshots showing almost complete reflection.

4.2 Partially blocked packet

The reference center $E_0 = 1.328814$ lies essentially on the second resonance peak, but the packet has a much larger finite momentum width $\Delta p = 0.015556$. Consequently only part of the incident spectrum sits in the high-transmission resonance region; neighboring components have much smaller $|T|^2$. Therefore, only the momentum components whose energies lie within the second resonance peak can be transmitted efficiently. Components whose energies deviate too far from the resonance have much smaller transmission probabilities and are reflected back to the left.

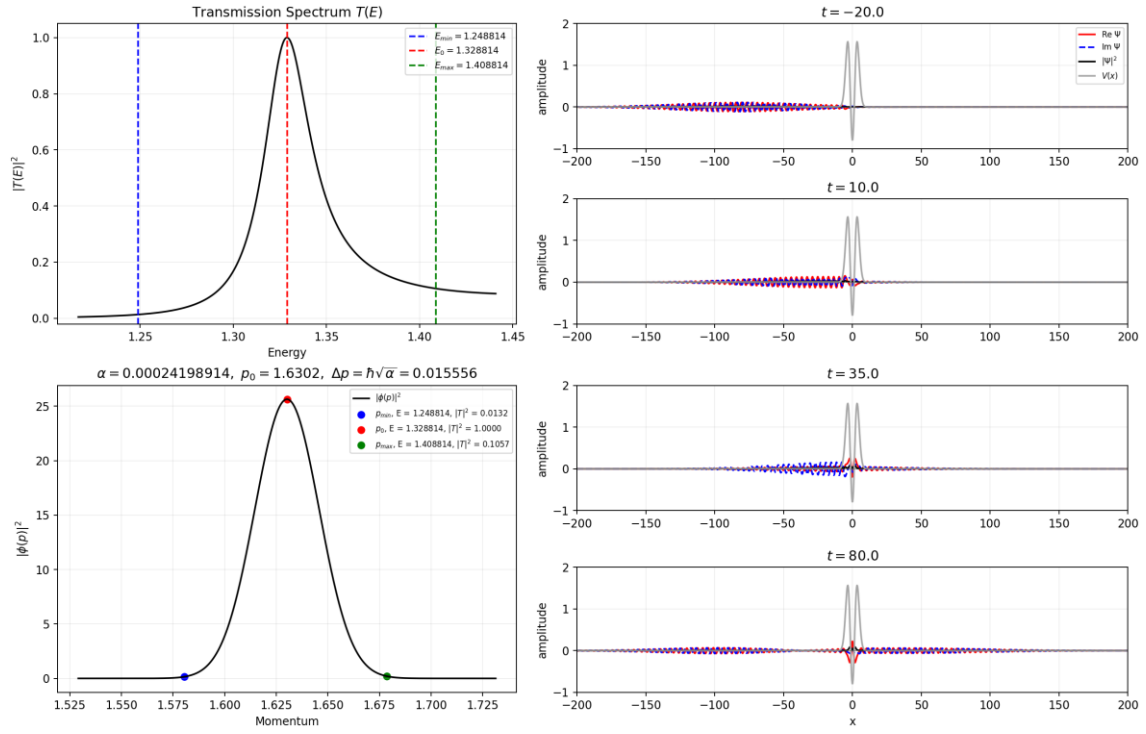


Figure 10. Partially blocked scenario centered near the second resonance. The finite spectral width covers both resonant and non-resonant energies, producing simultaneous transmission and reflection.

4.3 Passing packet

For $E_0=2.7$, the transmission probability is close to one over almost the whole momentum distribution. Therefore, almost all components of the wavepacket pass through the potential region and continue propagating to the right, with only a very small reflected part.

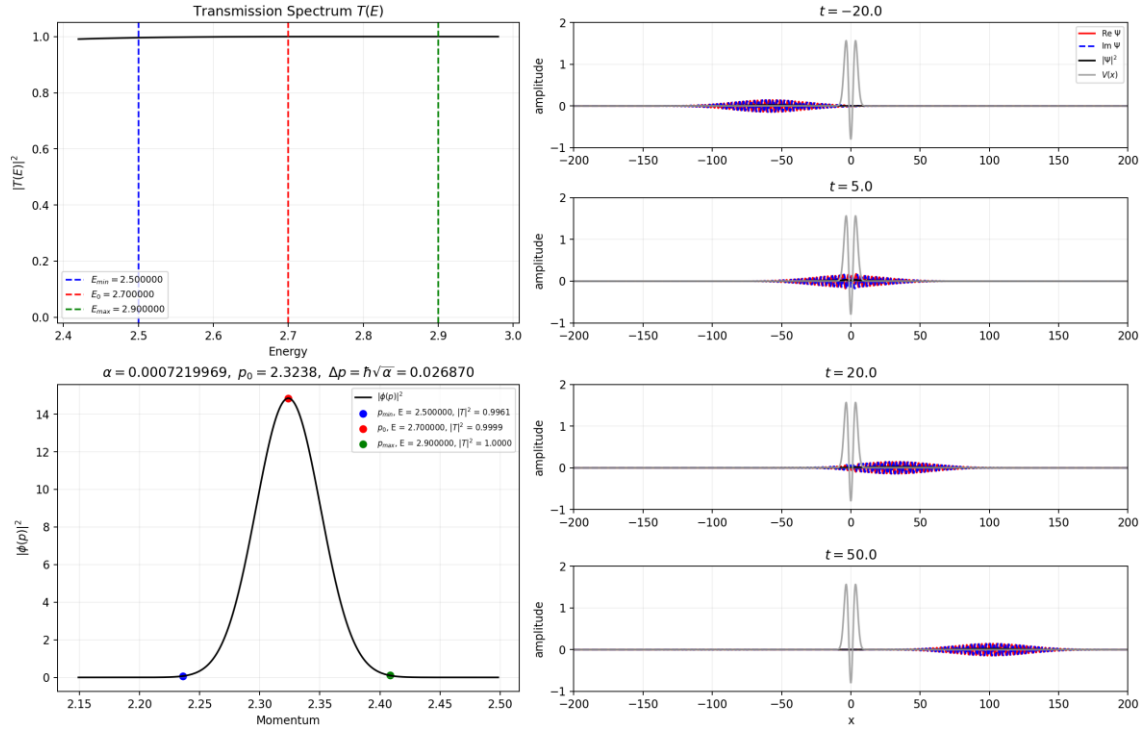


Figure 11. Passing scenario at high energy. Nearly the entire wavepacket crosses the potential.

4.4 Narrow range at first resonance peak

The first resonance at $E_0=0.62097030$ is considerably narrower than the second resonance, so $a_0=2.12 \times 10^{-6}$ is used to confine almost the whole incident spectrum to the high-transmission window. The momentum-space calculation gives $P_T=0.99836$ and $P_R=0.00164$; therefore, the packet is transmitted almost completely. This very small momentum width produces a real-space packet on the scale of 10^6 . In the full-range panels, the red real part and dashed blue imaginary part are retained over the complete propagation distance; the local animation then shows their rapid carrier oscillations around the potential.

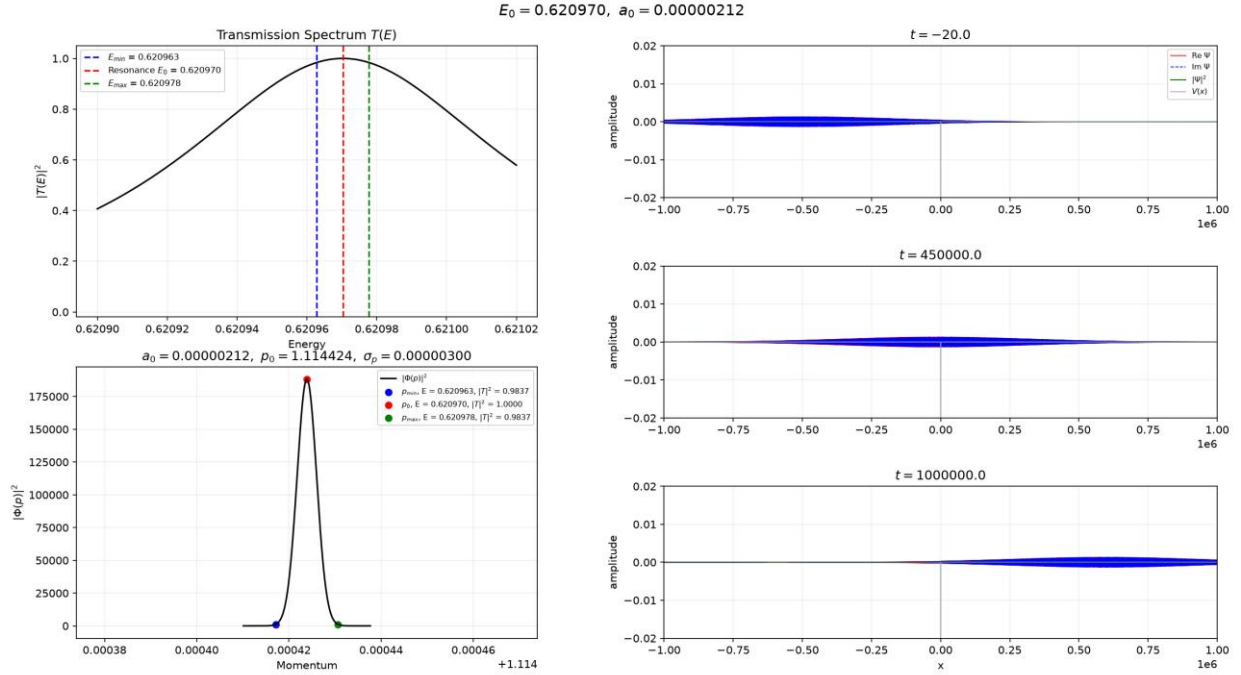


Figure 12. First-resonance packet centered at $E_0=0.62097030$. The three full-range snapshots show the real and imaginary parts before, during and after transmission.

4.5 Narrow range at second resonance peak

This case uses a very narrow momentum range centered on the second resonance peak. Since almost the entire range lies within the high-transmission region, the transmission probability is close to one for nearly all components of the wavepacket. Therefore, most of the wavepacket passes through the potential.

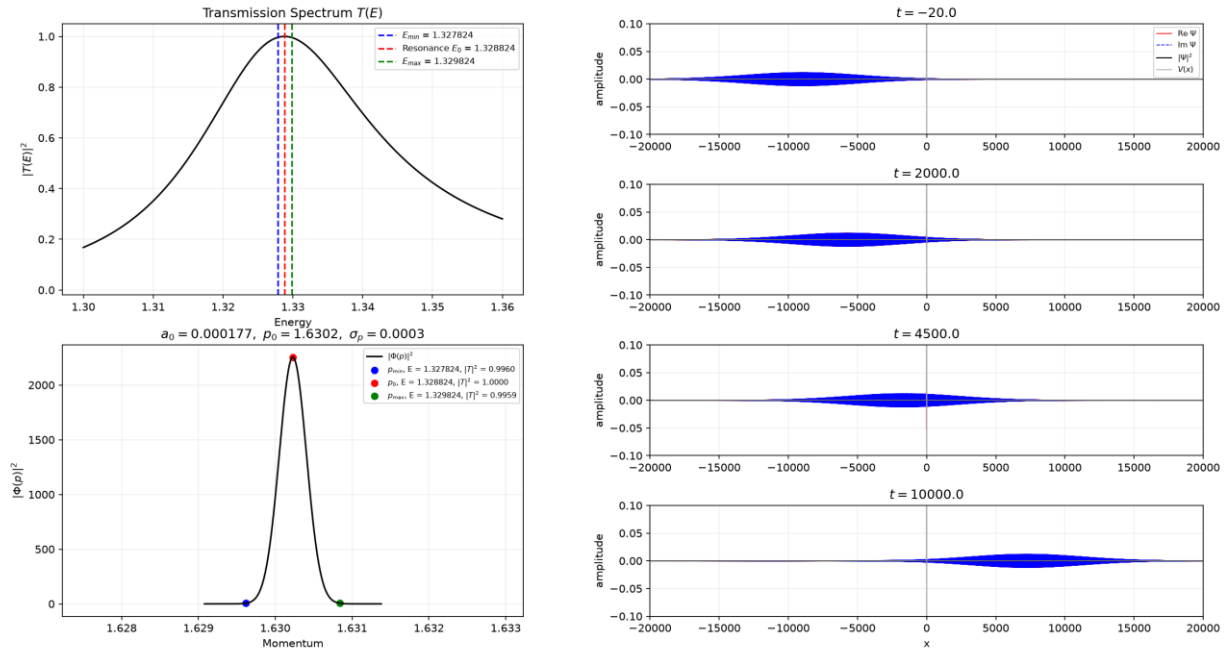


Figure 13. Extremely narrow-band packet centered exactly at the numerically determined second resonance. Its large spatial width is the Fourier counterpart of the tiny momentum width.